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Interface Circuits for a Six-Phase Variable Reluctance Energy Harvester

A comparative study of rectification and power conditioning
topologies

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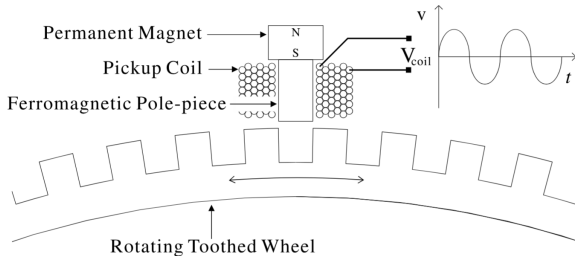
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Why energy harvesting?



- WSN and IoT nodes are everywhere, **collecting data** for many different purposes.
- Batteries are costly to maintain, have a finite life and are often **impossible to replace**, and they are bad for the environment.
- With **energy harvesting** the node powers itself from the energy already available around it.



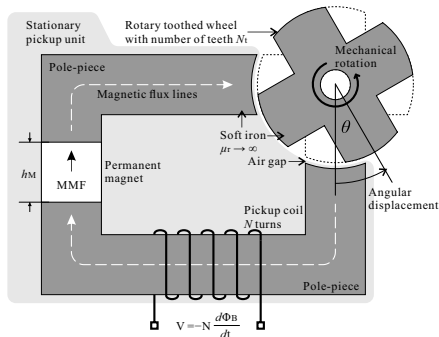
The variable reluctance principle



- Magnet and coil **stand still**; only the toothed wheel rotates.
- The passing teeth modulate the flux and induce an EMF.

Advantages

- No moving electrical parts
- Scalable to any shaft size
- Very small counter torque



Pickup unit facing the rotating wheel.

The six-phase MP-VREH

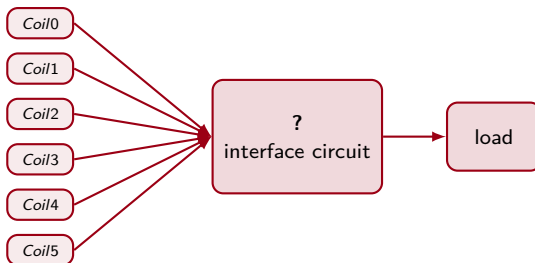


Six pickup units around the shaft, electrically 60° apart, evolved from railway prototypes into a design for large vehicles.

rpm	f (Hz)	V_{pk} (V)	R_{coil} (Ω)	L_{coil} (mH)	P_o (W)
100	36.7	1.82	6	21.68	0.41
200	73.3	2.7	6	21.46	0.91
300	110.0	3.18	6	21.27	1.26
400	146.7	3.48	6	21.12	1.51

Target range **100 - 400 rpm** (20 - 80 km/h), with under 2V peak per phase, so the rectifier choice becomes critical.

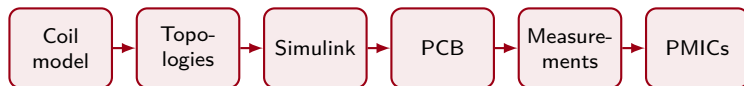
six coils, 60° apart



More phases, more power, but **six AC signals** to rectify, and prior work compared rectifiers for a **single phase** only.

Research question

Which interface circuit extracts the most power, over the whole speed range?

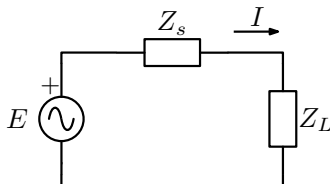
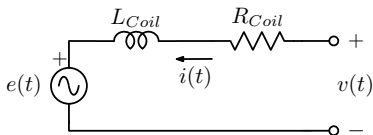


From theory to a real prototype, with **every topology measured on the same board** and on the same test bench.

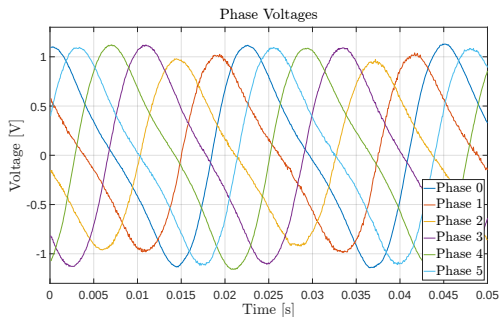
- Each coil is a source in series with R_{Coil} and L_{Coil} .
- At low speed a **resistive** load is enough.
- At higher speed $\omega_e L_{\text{Coil}}$ breaks the matching.
- A series capacitor C_m cancels it

$$C_m = \frac{1}{\omega_e^2 L_{\text{Coil}}}$$

valid at one frequency at a time.

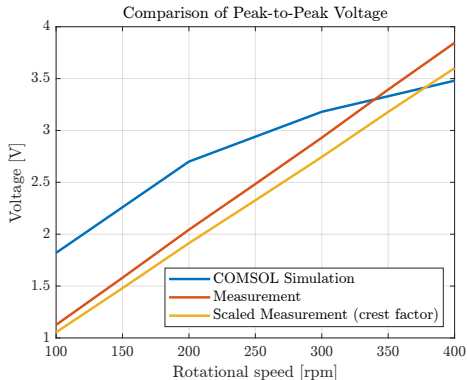


The six real phase voltages



- Measured at 100 rpm.
- The 60° shift is clearly visible.
- **Not** clean sine waves, and the six units are not identical.

- RMS voltage computed against speed.
- Each source replaced by an **equivalent sinusoid**, scaled by the crest factor.
- Same power as the real waveform.



FEM, raw measured peak, crest-factor corrected.

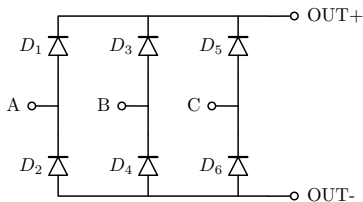
Strategy 1 — FWR bridges



Two three-phase groups ($k = 0, 2, 4$ and $k = 1, 3, 5$), each on a standard bridge.

- The **star (Y)** gives a line voltage $\sqrt{3}$ times the phase EMF, so the two-diode drop costs less.
- The **delta (Δ)** gives the coil EMF itself, with no gain but no floating node.

The star should give the **higher open-circuit voltage**.



Two diodes always in the current path.

Strategy 2 — NVC bridges

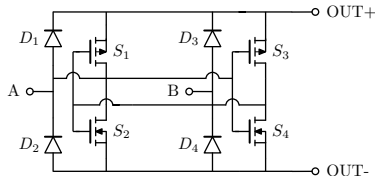


Three two-phase groups, pairing phases 180° apart in **anti-series**

$$v_A - v_B = e_k - e_{k+3} = 2 e_k$$

ideally double the voltage.

- Four MOSFETs with **cross-coupled gates**, so no gate driver is needed.
- They bypass the diodes, trading the fixed drop for a small R_{on} .
- Below $V_{\text{off}} \approx 1.2 \text{ V}$ it falls back to the diodes.

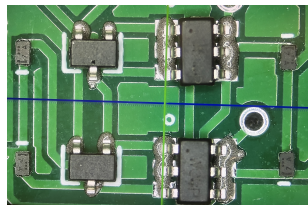
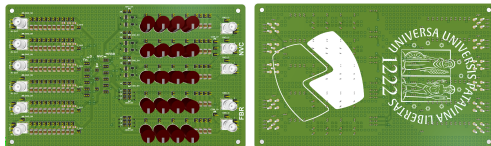


NVC for one anti-series pair.

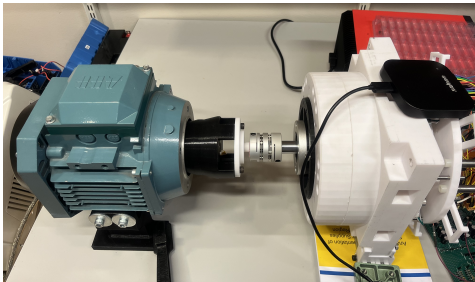
The prototype board



- Custom PCB, Altium Designer.
- Star, delta and NVC selected by moving jumpers, plus a jumper bank for C_m .
- Hand assembled with soldering iron and hot plate.



One NVC bridge during assembly.



Motor coupled to the MP-VREH.



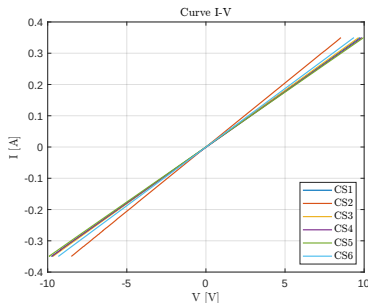
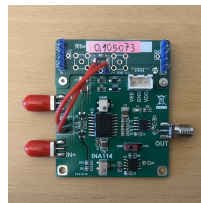
Acquisition device.



Programmable load.

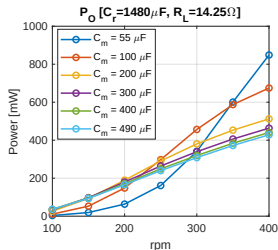
An inverter drives the motor and sets the speed precisely over the whole 100–400 rpm range.

- Six INA114AU boards, differential across a shunt.
- Shunts measured four-wire.
- Every sensor calibrated individually, into its own lookup table.
- Power trustworthy down to the milliwatt.

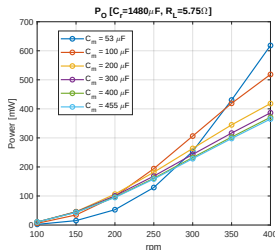


Calibration curves of the six sensors.

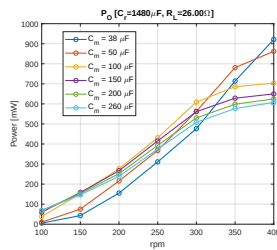
Simulation results



Star

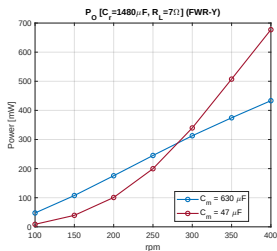


Delta

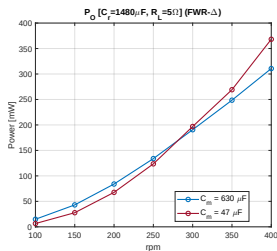


NVC

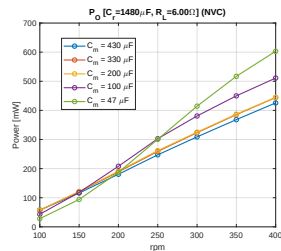
- Power against speed, for several fixed C_m .
- At 100 rpm, the worst case, the NVC gives **69.0 mW**, about $6\times$ the delta (10.7) and nearly double the star (36.7).
- The NVC is also the most efficient at every speed.



Star



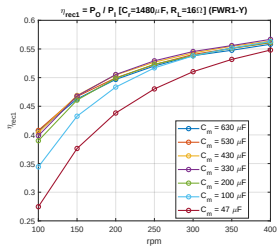
Delta



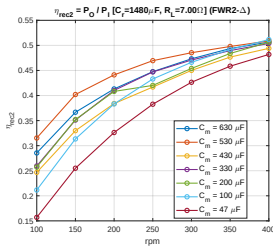
NVC

- At 100 rpm the NVC still wins with 57.7 mW, against 47.6 (star) and 14.9 (delta).
- Above 250 rpm the **star overtakes the NVC**, 433.3 against 425.7 mW at 400 rpm.
- Simulation did not predict this crossover.

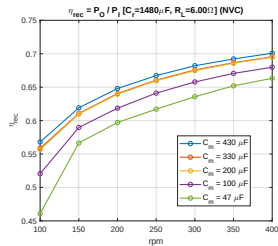
Measured efficiency



Star



Delta



NVC

- Best efficiency at every speed for the NVC, up to $\eta_{\text{Rec}} \approx 0.70$.
- Star ≈ 0.55 – 0.57 and delta only ≈ 0.48 – 0.51 , the weakest as expected.
- The two rankings disagree, since efficiency says NVC while power says star at high speed.

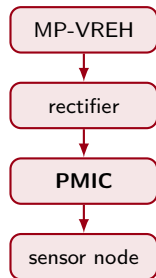
Measured power differs from simulation by 8% for the star, up to almost 30% for the NVC.

	Star	Delta	NVC
Mean deviation of P_O from simulation	8%	16%	30%

Why?

- The ceramic C_m lose capacitance under the real voltage, which grows with speed.
- Component tolerances and the parasitic resistance of the bridges.
- Sources that are not perfectly sinusoidal.

PMIC	V_{in} (V)	Star up to
DS-AEM13920_VR	0.120–4.6	250 rpm
DS-AEM10941_VR	0.05–5	250 rpm
bq25570	0.1–5.1	250 rpm
SPV1050 (Boost)	0.15–5.3	250 rpm
SPV1050 (Buck-Boost)	0.15–18	full range
SPV1040	0.3–5.5	300 rpm



The PMIC turns the rectified power into a stable supply.

- Delta and NVC stay below 4 V and are **compatible with every chip tested**.
- The star reaches 7 V and needs a wide-input converter or a clamp.

Conclusions

- **NVC and star** are the two strongest candidates, one per end of the speed range; delta is always the weakest.
- NVC and delta fit any PMIC tested; the star needs protection at high speed, its open-circuit voltage can damage the input.

Limitation

The harvester prototype is 3D-printed and sensitive to vibration at high speed, which makes the measurements harder to repeat.

Future work

Adaptive compensation tracking C_m with speed, and a full hardware test up to a real PMIC, to measure the efficiency from harvester to load.

Thank you for your attention!

Backup — star connection



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rpm	P_O (mW)		η_{Rec}	
	Simulated	Measured	Simulated	Measured
100	36.7	47.6	0.430	0.423
150	94.3	107.5	0.541	0.475
200	165.8	175.6	0.608	0.507
250	240.0	245.1	0.650	0.529
300	308.8	312.7	0.674	0.544
350	372.3	374.6	0.691	0.556
400	430.0	433.3	0.710	0.565

Backup — delta and NVC



Delta

rpm	P_O (mW)		η_{Rec}	
	Sim.	Meas.	Sim.	Meas.
100	10.7	14.9	0.210	0.279
150	43.7	43.2	0.340	0.379
200	95.0	84.2	0.425	0.441
250	158.3	134.1	0.482	0.480
300	228.5	190.8	0.522	0.508
350	298.0	248.7	0.550	0.515
400	365.0	310.7	0.570	0.511

NVC

rpm	P_O (mW)		η_{Rec}	
	Sim.	Meas.	Sim.	Meas.
100	69.0	57.7	0.665	0.568
150	146.5	109.4	0.768	0.619
200	238.9	175.9	0.821	0.648
250	374.8	245.2	0.841	0.668
300	508.0	308.0	0.842	0.682
350	576.8	367.7	0.834	0.693
400	610.0	425.7	0.820	0.701

rpm	Star		Delta		NVC
	V_{OC} (V)	ripple	V_{OC} (V)	ripple	V_{OC} (V)
100	1.98	0.012	1.000	0.022	2.70
150	2.83	0.012	1.460	0.021	3.05
200	3.67	0.011	1.910	0.017	3.20
250	4.51	0.012	2.365	0.017	3.15
300	5.32	0.012	2.825	0.017	3.15
350	6.17	0.011	3.275	0.017	3.15
400	7.04	0.012	3.755	0.017	3.15

The NVC saturates at 3.15 V above 200 rpm, with a ripple an order of magnitude larger than the FWR connections, consistent with its active switching operation.